

Original Article

Cerebral oximetry and postoperative delirium after cardiac surgery: a randomised, controlled trial*

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Summary

Postoperative delirium is associated with increased morbidity and mortality. We hypothesised that restoration of regional cerebral oxygen desaturation would reduce the incidence of postoperative delirium in elderly patients after cardiac surgery. After institutional ethics review board approval and informed consent, a double-blinded, prospective, randomised, controlled trial was conducted in patients ≥ 60 years of age undergoing cardiac surgery with cardiopulmonary bypass. In the intervention group, an algorithm was commenced if regional cerebral oxygen saturation decreased below 75% of baseline value for 1 min or longer. In the control group, the cerebral oximetry monitor screen was electronically blinded. Assessment of delirium was performed with confusion assessment method for intensive care unit or confusion assessment method after discharge from intensive care unit at 12-h intervals for seven postoperative days. Postoperative delirium was present in 30 out of 123 (24.4%) and 31 out of 126 (24.6%) patients in the intervention and control groups, respectively, odds ratio 0.98 (95%CI 0.55–1.76), $p = 0.97$. Postoperative delirium was present in 20 (71%) out of 28 and in 41 (18%) out of 221 patients with baseline regional cerebral oxygen saturation ≤ 50 , or $> 50\%$, respectively, $p = 0.0001$. Higher baseline regional cerebral oxygen saturation and body mass index were protective against postoperative delirium. Restoration of regional cerebral oxygen desaturation did not result in lower postoperative delirium after cardiac surgery. Pre-operative regional cerebral oxygen saturation $\leq 50\%$ was associated with increased postoperative delirium rates in elderly patients following cardiac surgery.

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Introduction

Cardiac surgery is associated with a significant risk of postoperative morbidity. Postoperative delirium (POD) may occur in up to one in three elderly patients

undergoing complex cardiac surgery [1]. It has been estimated that about \$6.9 billion US dollars of Medicare hospital expenditures are attributable to delirium [2]. The aetiology of delirium is complex, and includes

a variety of medical conditions and substances that interfere with brain function. The exact pathophysiological mechanism involved in the development of delirium following cardiac surgery is unclear. It is important to recognise that most of the independent predictors of delirium are difficult to modify, so that preventive strategies that reduce the incidence and severity of delirium should be prioritised.

One such strategy might include the peri-operative use of near-infrared reflectance spectroscopy (NIRS), which measures regional cerebral oxygen saturation (rScO₂) and detects real-time cerebral ischaemia. Interventions designed to minimise peri-operative reductions in rScO₂ may improve overall outcomes, particularly if rScO₂ is considered a reflection of adequacy of global perfusion, using the brain as an index organ [3]. This is of particular importance in emphasising the principle that if the brain is 'not OK', then be aware and beware [4].

Low rScO₂ values have been associated with post-operative cognitive dysfunction [5, 6], peri-operative stroke [7], increased incidence of major organ morbidity [8] and even 30-day and 1-year mortality after cardiac surgery [9]. Although the debate about whether all cardiac surgical patients should have intra-operative cerebral oxygenation monitoring is ongoing [10, 11], the need for better clinical evidence for NIRS technology is clearly recognised [12, 13]. Currently, the recommendations of the Association of Anaesthetists of Great Britain and Ireland for standards of monitoring during anaesthesia do not include cerebral oximetry monitoring as a standard of care [14].

We describe a prospective randomised clinical trial that assessed the effectiveness of peri-operative cerebral oximetry monitoring in reducing delirium following cardiac surgery. We hypothesised that restoration of rScO₂ desaturation to baseline values would reduce the incidence of delirium in elderly patients.

Methods

After Research Ethics Board approval at University Health Network, Toronto, Ontario, Canada, we conducted a prospective randomised, double-blind, controlled clinical trial from January 2012 to April 2015. This was a single-centre study in the quaternary referral hospital in Toronto, Canada.

The study group was patients ≥ 60 years of age, undergoing combined valve and coronary revascularisation procedures, repeat cardiac surgery, multiple valve replacement or repair, or surgery of ascending aorta and aortic arch, with or without circulatory arrest. We recruited patients through the pre-admission clinic and they gave written, informed consent. Patients with a history of serious mental illness, delirium or who were planned to undergo either emergency or surgery without bypass were not included in the study.

Anaesthesia management was conducted according to routine clinical practice. Premedication with oral lorazepam 1–2 mg was optional. The use of midazolam during surgery was limited to a maximum of 0.05 mg.kg⁻¹. Anaesthesia was induced with fentanyl 10–12 µg.kg⁻¹, propofol 0.5–2 mg.kg⁻¹ and rocuronium 0.6 mg.kg⁻¹, and maintained with 0.5–2.0% isoflurane. Heart rate and blood pressure were kept within 25% of baseline values. Heparin was administered to maintain an activated clotting time above 480 s.

The cardiopulmonary bypass circuit was primed with 1.8 l of Ringer's lactate and 50 ml of 20% mannitol. Management of bypass included systemic temperature drift to 34 °C, alpha-stat pH management, targeted mean perfusion pressure between 60 mmHg and 80 mmHg and pump flow rates of 2.0–2.4 l.m⁻².min⁻¹. Myocardial protection was achieved with intermittent antegrade, and occasionally retrograde blood cardioplegia. A 32-µm filter (Avecor Affinity, Minneapolis, MN, USA) was used in the arterial perfusion line.

Deep hypothermic circulatory arrest was achieved by cooling to 20 °C with antegrade cerebral perfusion. Before separation from CPB, patients were rewarmed to 36–37 °C. During rewarming, the maximal inflow temperature was limited to 37 °C. After separation from bypass, heparin was neutralised with protamine sulphate (1 mg per 100 U of heparin) to achieve an activated clotting time within 10% of baseline. All patients were transferred to the intensive care unit (ICU) after surgery.

Postoperative sedation was achieved with propofol infusion 25–50 µg.kg⁻¹.min⁻¹ until tracheal extubation criteria were met. Both groups received a combination

of opioid analgesics and non-opioid adjuvants for postoperative pain management.

This was a superiority study, hypothesising that restoration of rScO₂ would result in lower POD rates after cardiac surgery. Before surgery, patients were randomly allocated to either interventional or control groups according to a computer-generated randomisation code in blocks of four, aiming at subject allocation in a 1:1 ratio.

Bilateral NIRS sensors (INVOS™ 5100C; Covidien) were used to measure rScO₂ intra-operatively and postoperatively, up to 24 h in ICU. The sensors were placed on the fronto-temporal area, and baseline values of rScO₂ were obtained (according to manufacturer's guidelines) in the operating room before induction of anaesthesia.

In the intervention group, an alarm threshold at 75% of the baseline rScO₂ value (25% reduction from baseline) was established. Interventions were commenced if the rScO₂ value decreased below 75% of baseline for 1 min or longer. A similar algorithm to that which was previously reported by Murkin et al. [8] was utilised. It included potential mechanical and physiological causes of low rScO₂ values: checking head position and venous cannula position to ensure adequate venous drainage; maintaining jugular venous pressure < 10 mmHg by repositioning the heart during bypass; maintaining cerebral perfusion pressure > 50 mmHg; increasing FiO₂ to 1.0; increasing depth of anaesthesia to decrease cerebral oxygen consumption; optimising PaCO₂ (maintain PaCO₂ close to 5.3 kPa); maintaining mean arterial pressure ≥ 50 mmHg; transfusing red blood cells if haematocrit is < 20%; increasing cardiac output if cardiac index is < 2.0 l. m⁻².min⁻¹; facilitating cooling if core temperature exceeds 38.5 °C. If seizures were present, aggressive anticonvulsant therapy was commenced. There was no stepwise strategy to restore rScO₂ value; more than one intervention was allowed at any one time. However, the actual target treatment physiological parameters for individual patients were not quantified.

In the control group, the cerebral oximetry monitor screen was electronically blinded. However, recording was continuous following verification of the signal strength and baseline value by an independent observer trained in cerebral oximetry application, who was

also unaware of the study design. The rScO₂ data for both groups were stored on a dedicated computer for consequent analysis.

Baseline and minimum values of rScO₂, number of desaturation events, area under the curve (AUC) of rScO₂ values below 75% of baseline (AUC rScO₂ < 75% baseline), as well as the effectiveness of restoration of rScO₂ were determined. The AUC included both the duration and severity of rScO₂ desaturations (based on 25% desaturation from baseline), and was calculated by using INVOS analytic tool software (provided by Covidien Canada Ltd, Mansfield, MA, USA).

Assessment of delirium was performed pre-operatively (baseline) and postoperatively at 12-h intervals, or as needed (if the patient's condition changed or antipsychotic medications were administered), using the confusion assessment method for ICU (CAM-ICU) [15]. When patients were discharged from ICU to the surgical floor, delirium was assessed using the confusion assessment method (CAM). Patients were assessed for delirium for the first seven postoperative days or until discharge. Patients were considered delirious until they were deemed CAM-negative. The confusion assessment method was used for both ventilated and extubated patients. It included a four-step algorithm to identify the following: (1) acute onset of changes or fluctuations in the course of mental status; (2) inattention; (3) disorganised thinking; (4) altered level of consciousness. Patients were considered delirious if both features (1) and (2) were present, plus either feature (3) or (4). Patients were recorded as either CAM-positive (delirium present) or CAM-negative (delirium absent). The CAM-ICU and CAM testers were research assistants who were trained and had been performing these tests for a number of years. They were not aware of the study objectives. Intravenous haloperidol was used as first-line treatment in delirious patients, given in increments of 1–5 mg, repeated every 30–60 min as needed. Other antipsychotic medications were administered as required.

Major adverse outcomes were recorded. These included: stroke (functional deficit exceeding a 24-h period); myocardial infarction (troponin assays); renal failure (dialysis); pneumonia (clinical, X-ray); atrial fibrillation (documented electrocardiogram); sepsis

(microbiology laboratory confirmation, septic shock); systemic inflammatory response syndrome (vaso-active medication requirement); and unplanned return to the operating room. Duration of ICU and hospital length of stay, and all-cause in-hospital mortality were also recorded.

Given the 30% prevalence of POD after cardiac surgery, to detect a 50% reduction of POD in patients receiving rScO₂ restoration management strategy with $\alpha = 0.05$ and power $1 - \beta = 0.8$, a group of 121 patients in each arm of the study are required, making a total of 242 patients in the randomisation schedule.

Descriptive analysis was performed for all variables measured before and after surgery. Continuous data were presented as mean (SD), or median (IQR [range]). Dichotomous data were presented as n (%).

For the primary outcome of delirium, the two groups were compared with the Chi-square test for differences in proportions in a 2×2 contingency table. Odds ratio and confidence intervals for proportions were calculated at 95%. A two-tailed Student's t-test was used for two independent samples to analyse continuous normally distributed data. The Mann–Whitney U-test was used for nonparametric data. A $p < 0.05$ was considered statistically significant. All analyses were performed on an intention-to-treat basis.

To determine the predictors of POD, a post-hoc logistic regression model using backwards stepwise methodology was used. The threshold for inclusion in the model was selected at $p < 0.10$. Initial covariates included in the model were: left and right baseline rScO₂ measures; log¹⁰ of combined desaturation AUC (because area values ranged from 0 to over 12,500); dichotomous (yes or no) desaturation covariate; age; sex; categorical body mass index ((BMI); normal, overweight, obese); history of pre-operative cerebrovascular accident; history of pre-operative diabetes mellitus; hypertension; bypass duration; alcohol use; smoking status; pre-operative use of beta-blockers and statins; number of units of red cells transfused; overall transfusion (dichotomous variable, yes or no); and a covariate for any blood product used.

The Hosmer–Lemeshow goodness-of-fit test was used to assess the fit of the model, while performance of the model was assessed using the receiver operating characteristic (ROC) curve and the associated c-

statistic. The area under the ROC curve for the reduced model was compared with the area under the ROC curve for the base model (with all initial covariates included), to test whether backward stepwise covariate elimination affected the model performance. Statistical analysis was performed using STATA 12.1 (StataCorp, College Station, TX, USA).

Results

A total of 1248 subjects were screened for eligibility, and 998 were not studied due to patient refusal, not meeting the criteria, language barrier or prior recruitment to other studies. A total of 250 patients were randomly allocated, and 249 analysed. One patient from the intervention group died in the operating room. As a result, there were a total of 123 and 126 patients in the intervention and control groups, respectively (Fig. 1). Both groups were similar with respect to baseline characteristics, pre-operative medications, comorbidities and surgical characteristics (Table 1).

Postoperative delirium was present in 30 out of 123 (24.4%), and 31 out of 126 (24.6%) patients in the intervention and control groups, respectively (OR 0.98, 95% CI 0.55–1.76, $p = 0.97$). A total of 24 (80%) out of 30, and 26 (84%) out of 31 patients in the intervention and control groups developed POD while still in ICU. The remaining six (20%) and five (16%) patients in their respective groups developed POD after ICU discharge. Postoperative delirium rates were similar in patients with and without cerebral desaturations, as well as in patients with restored and not restored rScO₂ (Fig. 2).

There were no significant differences between the two groups with respect to incidence or duration of cerebral desaturations (Table 2). During the entire study period, a total of 287 and 245 desaturation episodes (lasting ≥ 1 min) occurred in the intervention and control groups, respectively. Thirty-eight percent vs. 40% of these desaturation episodes were recorded in the operating room, and 62% vs. 60% in ICU, in the intervention and control groups, respectively. In the operating room, the majority of desaturations happened during bypass in both groups (Fig. 3). The success rates in restoring rScO₂ to baseline values differed significantly between attempts in the operating room and the ICU. Restoration of rScO₂ was achieved more consistently in the operating room compared with the

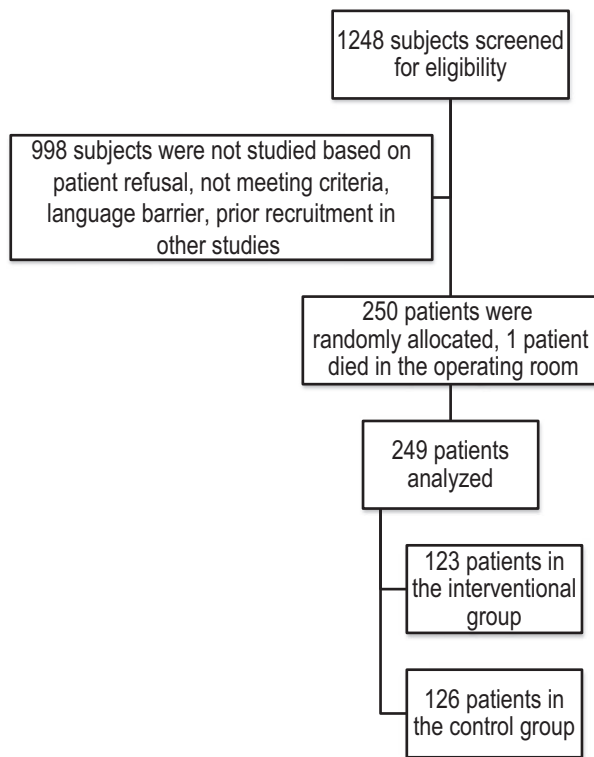


Figure 1 CONSORT diagram.

ICU (Fig. 4). The number of different therapeutic measures to restore rScO₂ are described in Table 3.

Baseline rScO₂ ≤ 50% was present in 28 patients (15 interventional group and 13 controls). Postoperative delirium was present in 20 (71%) out of 28 patients (9 interventional group and 11 controls) with baseline rScO₂ ≤ 50%, compared with 41 (18%) out of 221 patients with baseline rScO₂ > 50% (OR 3.8, 95% CI 1.98–7.47, *p* = 0.0001).

Haloperidol was administered in 12 and 14 patients in the intervention and control groups, respectively. Furthermore, quetiapine was administered in four and five patients in their respective groups. The overall incidence of major adverse outcomes, requirements for inotropic/vasoconstrictor support and the length of stay was similar in the two groups (Table 4).

When the logistic regression model was utilised with significance threshold for inclusion of *p* < 0.10, 12 out of the initial 17 covariates were removed from the model, with right baseline rScO₂, BMI, history of stroke, bypass duration and smoking status remaining (Table 5). Higher baseline rScO₂ was protective against POD, whereas longer bypass duration was a risk factor

for POD. Interestingly, overweight or obese patients had a lower incidence of delirium. The logistic model was a good fit, based on a non-significant Hosmer–Lemeshow statistic (*p* = 0.34), and performed well with a ROC area of 0.79. Lastly, when the ROC area of this reduced model (0.79) was compared with the ROC area of the full model with all initial covariates (0.81) there was no significant difference (*p* = 0.09), suggesting no difference in model performance due to the backwards stepwise elimination of non-significant covariates.

Discussion

This was a large-scale randomised, controlled clinical trial that examined the effects of peri-operative surveillance of rScO₂ and restoration of low rScO₂ on the incidence of POD in elderly patients after cardiac surgery. We found no difference in the incidence of POD between the intervention group (algorithm designed to restore low rScO₂) and control (no interventions) group. Postoperative delirium rates were similar in patients with and without rScO₂ desaturations, regardless of the group assignment. Even though the median duration of cerebral desaturation (AUC %_{min}) was 2.2-fold higher in the control group than the intervention group (Table 2), this was not a statistically significant difference. It is likely that, with the wide range in AUC %_{min} a much larger sample size would be required to achieve statistical significance.

It is important to note, however, that there was a four-fold increase in POD rates in patients with baseline rScO₂ ≤ 50% when compared with patients with rScO₂ > 50%. This finding supports the previous report by Schoen et al. [16] who conducted an observational study of 231 patients, showing that pre-operative rScO₂ < 59.5% was predictive of POD after elective cardiac surgery. Additionally, Heringlake et al. [9] reported that pre-operative rScO₂ ≤ 50% was an independent risk factor for 1-year mortality in a large cohort of 1178 patients undergoing cardiac surgery, independent of the severity of pre-operative EuroSCORE.

Several other studies have reported a correlation between low rScO₂ and cognitive decline after cardiac surgery. Yao et al. [5] showed that intra-operative rScO₂ < 40% was predictive of postoperative cognitive dysfunction as measured by the Mini-Mental State

Table 1 Baseline characteristics and surgical characteristics of study population. Values are mean (SD), number (proportion), mean (SD) or median (IQR [range]).

	Intervention group n = 123	Control group n = 126
Age; years	74.2 (6.5)	72.9 (6.3)
Sex; male	88 (71.5%)	88 (69.8%)
BMI; kg.m ⁻²	27.6 (4.8)	28.5 (5.1)
Comorbidities		
Coronary artery disease	88 (71.5%)	79 (62.7%)
Hypertension	90 (73.2%)	101 (80.1%)
Myocardial infarction	16 (13.0%)	15 (11.9%)
Atrial fibrillation	31 (25.2%)	42 (33.3%)
Congestive heart failure	24 (19.5%)	28 (22.2%)
Stroke/Transient ischaemic attack	21 (17.0%)	15 (11.9%)
Diabetes mellitus	36 (29.3%)	33 (26.2%)
Peripheral vascular disease	9 (7.3%)	11 (8.7%)
Thyroid disease	18 (14.6%)	18 (14.2%)
COPD	17 (13.8%)	15 (11.9%)
Left ventricular grade	1 (1–2 [1–4])	1 (1–2 [1–4])
Pre-operative medications		
Beta-blockers	67 (54.5%)	69 (54.7%)
Calcium channel blockers	33 (26.8%)	34 (26.9%)
ACE inhibitors	41 (33.3%)	51 (40.5%)
Statins	78 (63.4%)	86 (68.2%)
Aspirin	81 (65.8%)	84 (66.6%)
Antidepressants	7 (5.7%)	11 (8.7%)
Benzodiazepines	9 (7.3%)	14 (11.1%)
Lorazepam premedication	60 (48.8%)	66 (52.3%)
Alcohol intake	54 (44%)	47 (37%)
Units per week	0 (0–7 [0–55])	0 (0–3 [0–60])
Smoking	36 (29%)	42 (33%)
Haemoglobin; g.l ⁻¹	134.2 (15.3)	135.7 (16.9)
Creatinine; µmol.l ⁻¹	96.3 (30.4)	92.3 (24.3)
Type of surgery		
CABG	65 (52.8%)	59 (46.8%)
Mitral valve	33 (26.8%)	37 (29.3%)
Aortic valve	63 (51.2%)	70 (55.5%)
Ascending aorta/Arch	24 (19.5%)	30 (23.8%)
Redo sternotomy	12 (9.7%)	10 (7.9%)
DHCA	11 (8.9%)	14 (11.1%)
Any two valves	19 (15.4%)	16 (12.7%)
Any three valves	11 (8.9%)	9 (7.1%)
CABG plus valve(s)	53 (43%)	49 (38.9%)
Redo-CABG	9 (7.3%)	6 (4.8%)
Redo-CABG plus valve	3 (2.4%)	4 (3.2%)
Distal anastomoses	1 (0–2 [0–5])	1 (0–2 [0–5])
CPB time; min	108 (83–136 [34–382])	114 (85–148 [52–336])
Cross-clamp time; min	85 (64–109 [27–282])	91 (63–123 [28–193])

COPD, chronic obstructive pulmonary disease; ACE, angiotensin-converting enzyme; CPB, cardiopulmonary bypass, CABG, coronary artery bypass graft; DHCA, deep hypothermic circulatory arrest; Left ventricular grading, 1 (ejection fraction > 60%), 2 (ejection fraction 40–59%), 3 (ejection fraction 20–39%), 4 (ejection fraction < 20%).

Examination and the antisaccadic eye movement tests. Slater et al. [6] found that patients with significant rScO₂ desaturations during primary coronary revascularisation surgery had a significantly higher risk of early postoperative cognitive decline, as well as prolonged hospital stay. De Tournay-Jetté et al. [17]

reported that an absolute value of rScO₂ < 50%, as well as the relative decrease in rScO₂ by more than 30% were associated with early and late cognitive decline after bypass. A more recent study by Colak et al. [18] demonstrated that prolonged rScO₂ desaturation was a significant predictor of cognitive decline

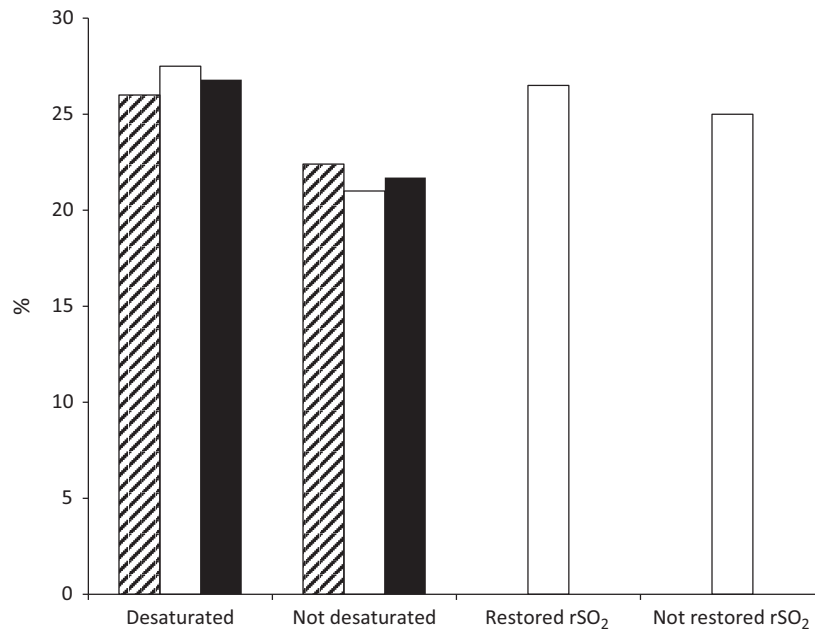


Figure 2 Postoperative delirium in different sub-groups of patients. All patients (▨); Intervention group (□); Control group (■).

Table 2 Prevalence of cerebral desaturations in operating room and ICU. Values are mean (SD), number (proportion) or median (IQR [range]).

	Intervention group n = 123	Control group n = 126
Baseline rScO ₂ ; %		
Right side	63.1 (10.4)	62.7 (10.5)
Left side	63.4 (9.8)	62.3 (10.3)
Incidence of cerebral desaturations		
During entire study period	65 (52.8%)	69 (55%)
Only OR	15 (12%)	21 (17%)
Only ICU	21 (17%)	25 (20%)
Both, OR/ICU	29 (23%)	23 (18%)
Number of desaturation episodes		
OR	1 (0–3 [0–9])	1 (0–2 [0–10])
ICU	2 (1–4 [0–14])	1 (0–3 [0–16])
Both, OR/ICU	3 (1.7–7.0 [1–19])	2 (1–4 [1–17])
Duration of cerebral desaturation in OR; min		
All patients	11.2 (4.4–35.8 [1.9–146.3])	20.3 (4.8–52.7 [1.2–274.8])
No algorithm employed (n = 11)	5.73 (2.9–5.8 [1.9–7.5])	
rScO ₂ restored with algorithm (n = 24)	11.7 (6.7–20.5 [7.5–28.3])	
rScO ₂ not restored with algorithm (n = 9)	73.8 (40.9–90.2 [37.4–146.3])	
Duration of cerebral desaturation in ICU; min		
All patients	29.9 (17.2–108.9 [1.3–1135.1])	57.3 (12.1–226.8 [2.1–1151.3])
No algorithm employed (n = 34)	22.8 (8.5–57.5 [1.3–154.2])	
rScO ₂ restored with algorithm (n = 10)	31.7 (21.7–69.4 [20.4–122.0])	
rScO ₂ not restored with algorithm (n = 6)	386.1 (267.2–883.3 [154.2–1135.1])	
Combined OR/ICU AUC rScO ₂ < 75% baseline	36 (14–702 [2–12 520])	80 (22–301 [2–10 831])

OR, operating room; ICU, intensive care unit; rScO₂, regional cerebral oxygen saturation. Cerebral desaturation was defined as a reduction in rScO₂ value by 25% or more from the baseline value. If the rScO₂ value was not restored to baseline, the patient was classed as 'not restored with algorithm'.

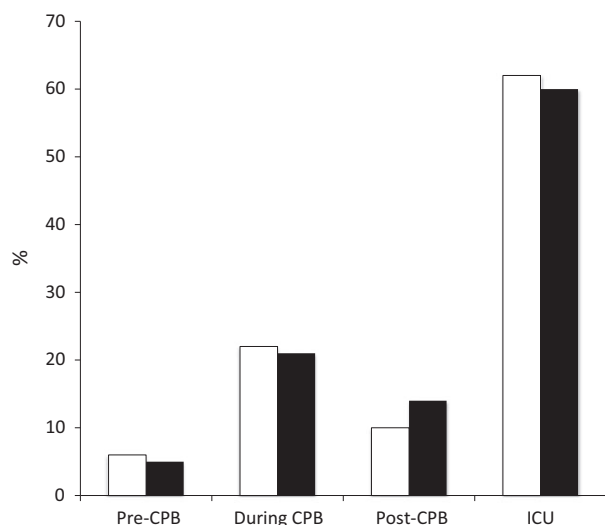


Figure 3 Prevalence of cerebral desaturation during the study period. CPB, cardiopulmonary bypass. Intervention group (□); Control group (■).

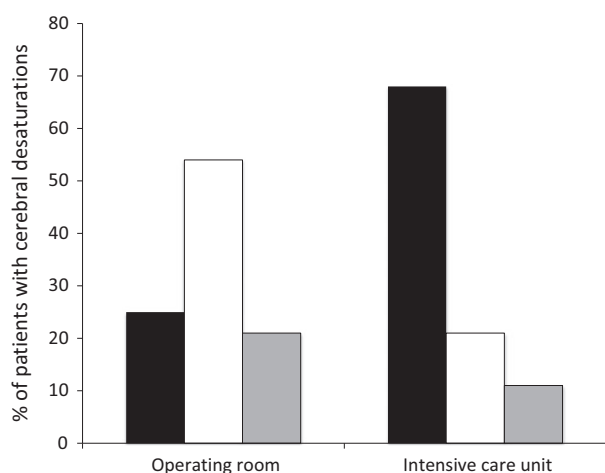


Figure 4 Cerebral regional oxygen saturation (rScO₂) restoration rates in the intervention group. No Intervention group (■); rScO₂ restored (□); rScO₂ not restored (■).

Table 3 The list of therapeutic measures to restore cerebral desaturations.

	Number of events
Increase in mean arterial pressure	126
Increase in FiO ₂	31
Check head and venous cannula position	27
Adjust depth of anaesthesia	21
Increase haemoglobin concentration	17
Adjust PaCO ₂	14
Increase cardiac index	12

after cardiac surgery with bypass, and that maintaining rScO₂ > 80% of baseline could potentially reduce this incidence.

Two recent multi-centre pilot and feasibility studies were conducted to determine the frequency of rScO₂ desaturations during bypass, and the effectiveness of interventions designed to reverse them. Subramanian et al. [19] defined cerebral desaturation as a reduction in rScO₂ of more than 20% from baseline, which was present in 143 (61%) of 235 patients during bypass. It is interesting to note that spontaneous resolution of desaturation was present in 115 (33%) out of 380 desaturation events, but the application of a treatment algorithm resulted in reversibility of rScO₂ in all but 5% of patients. Deschamps et al. [20] initiated an algorithm for reversal of rScO₂ desaturation using a 10% reduction from the baseline rule. This study was a feasibility trial including 201 patients from eight Canadian centres, achieving 97% reversal of intra-operative cerebral desaturations. The pivotal difference between the NORMOSAT study [20] and this study was the treatment thresholds for cerebral desaturation, that is, 10% vs. 25% reduction of rScO₂ below baseline. It is likely that earlier interventions in the NORMOSAT study [20] resulted in higher success rates for rScO₂ restoration.

In this study, we found that standard clinical care in the operating room was associated with reversal of rScO₂ in one out of four patients in whom the study algorithm was not employed. The duration of rScO₂ desaturations in these patients was 5.7 (1.9–7.5) minutes. Subramanian et al. [19] reported that spontaneous rScO₂ restoration was noted in 30% of patients. Deschamps et al. [20] intervened within 15 s of 10% rScO₂ reduction from baseline, and achieved a reversal rate of 97%. It is possible that keeping the rScO₂ very close to baseline at all times would be a better management strategy. Different studies have used different definitions (10–25%) with respect to the threshold of rScO₂ reduction from baseline necessitating intervention. Similarly, successful reversal rates in the same studies have ranged from 80% to 97% [8, 19, 20].

Spontaneous reversibility of rScO₂ in the ICU exceeded 20 min, to a maximum of 2.5 h in three patients. However, there was no difference in POD rates between patients with prolonged desaturations and no desaturations.

Table 4 Postoperative morbidity and mortality after cardiac surgery. Values are number (proportion) or median (IQR [range]).

	Intervention group n = 123	Control group n = 126
Outcome of interest		
Atrial fibrillation	73 (59%)	68 (54%)
Pneumonia	6 (5%)	2 (1.6%)
Sepsis	2 (1.6%)	2 (1.6%)
Chest reopening for bleeding	7 (6%)	8 (6%)
Dialysis	3 (2.4%)	1 (0.8%)
Stroke	4 (3%)	3 (2.4%)
Myocardial infarction	4 (3%)	4 (3%)
Any transfusion	46 (37%)	54 (43%)
Any inotropes	109 (89%)	110 (87%)
Death	3 (2.4%)	4 (3%)
Extubation time; h	7.2 (4.8–12.0 [0.2–538.8])	6.0 (4.2–12.9 [0.0–161.4])
Intensive care unit length of stay; h	49 (25–104 [4–813])	49 (23–95 [0–1316])
Hospital length of stay; days	8 (6–11 [5–67])	8 (6–11 [1–71])

Table 5 Multivariate analysis: Stepwise logistic regression analyses.

Covariates	Base logistic model (no covariate elimination)	p value	Logistic model, backwards stepwise elimination (p < 0.10)	p value
Baseline rScO ₂ (left)	0.99 (0.93–1.07)	0.83	–	
Baseline rScO ₂ (right)	0.94 (0.88–1.01)	0.09	0.93 (0.89–0.96)	< 0.0001
Desaturation (yes/no)	0.97 (0.26–3.62)	0.95	–	
Desaturation AUC, log ¹⁰	1.21 (0.63–2.33)	0.57	–	
Age; year	1.05 (0.99–1.12)	0.09	–	
Sex	1.55 (0.71–3.39)	0.27	–	
BMI; kg.m ⁻²				
< 25				
25–30	0.35 (0.14–0.84)	0.02	0.39 (0.17–0.87)	0.02
> 30	0.31 (0.11–0.86)	0.02	0.33 (0.14–0.81)	0.016
Past medical history				
CVA/TIA	2.07 (0.81–5.33)	0.13	2.13 (0.90–5.02)	0.08
Diabetes mellitus	1.65 (0.75–3.60)	0.21	–	
Hypertension	1.24 (0.49–3.15)	0.64	–	
Alcohol intake	1.01 (0.97–1.04)	0.65	–	
Smoking status	1.69 (0.74–3.86)	0.20	1.81 (0.90–3.65)	0.09
Beta-blockers	0.94 (0.47–1.90)	0.87	–	
Statins	0.86 (0.39–1.92)	0.71	–	
Cardiopulmonary bypass time; min	1.01 (1.00–1.02)	0.004	1.01 (1.00–1.02)	0.001
RBC transfusion	1.2 (0.94–1.74)	0.12	–	
Any transfusion (yes/no)	0.77 (0.30–1.95)	0.58	–	
ROC; c-statistic		0.81		0.79
Hosmer–Lemeshow (model fit)		0.83		0.34
ROC compared with base model		N/A		0.09

BMI, body mass index; CVA/TIA, cerebrovascular accident, transient ischaemic attack; RBC, red blood cells; AUC, area under the curve; ROC, receiver operating characteristic curve. Logistic regression results displayed as odds ratio (95% CI); N/A, not applicable.

It is likely that some of these rScO₂ desaturations reflected non-compliance with the protocol, but the majority of them were seen in patients with otherwise ‘acceptable’ physiologic limits. This finding was

particularly pertinent to a relatively short spontaneous recovery of rScO₂ in the operating room by applying standard clinical care. Most treatment algorithms are a combination of strategies to improve global cerebral

perfusion and oxygen delivery. The algorithm that was utilised in our study was based on correcting mechanical causes (e.g. head and cannula position), increasing oxygen supply (e.g. increasing FiO_2 , cardiac output, haemoglobin) and decreasing oxygen demand (e.g. decreasing body temperature, increasing anaesthesia depth). Although it is usually relatively easy to restore rScO_2 desaturations due to mechanical causes, the exact thresholds for different physiological parameters such as optimal mean arterial blood pressure, PaCO_2 , PaO_2 and haemoglobin correction for an individual patient are currently unknown. For example, should a patient receive red cell transfusion for haemoglobin level of 85 g.l^{-1} in ICU if rScO_2 is 25% below baseline, but all other physiological parameters are within the acceptable range? It is an open question that requires discussion. It is also important to note that recent evidence suggests that increasing mean arterial blood pressure by administration of phenylephrine or vasopressin may in fact reduce cerebral oxygenation as a result of cerebral vasoconstriction [21].

Furthermore, neither Subramanian et al. [19] nor Deschamps et al. [20] included patients undergoing deep circulatory arrest. Our study included patients undergoing aortic arch surgery with deep circulatory arrest, resulting in a considerable number of desaturations that were not amenable to interventions designed to restore rScO_2 .

Streamlining the current research developments in detecting lower and higher limits of cerebral autoregulation by employment of cerebral oximetry monitoring with integrated blood pressure variations may address some of the limitations of the current approach [22]. Furthermore, the exact threshold for the treatment of cerebral desaturation is not well established. We chose a 25% reduction from baseline as a threshold for interventions. Perhaps keeping values closer to the baseline would have been a better choice. It is possible that a larger study recruiting several thousands of patients may provide more information on the importance of rScO_2 reversibility with respect to clinically meaningful outcomes. It may also provide an indication of what cut-off limits for duration and severity of rScO_2 desaturations are associated with poor outcomes.

It was not surprising that longer bypass times were associated with higher risk of POD. Interestingly, we

also identified that being overweight (BMI $25\text{--}30 \text{ kg.m}^{-2}$) or obese (BMI $> 30 \text{ kg.m}^{-2}$) seemed to confer protection against POD. It has been reported that BMI of $25\text{--}30 \text{ kg.m}^{-2}$ is associated with lower mortality after cardiac surgery [23], and the BMI $35\text{--}40 \text{ kg.m}^{-2}$ resulted in lower incidence of peri-operative stroke after non-cardiac surgery [24]. Although intriguing, these incidental findings require further investigation.

The current study has several limitations. Restoration of rScO_2 occurred without any documented interventions in 25% of patients in the operating room, and 68% of patients in ICU. This could have been related to non-compliance with the protocol, possibly due to conflicting clinical priorities, or it is possible that the algorithm was followed but the interventions were not entered into the monitor, that is, not recorded. Another possibility is that the physiological limits could have been within an acceptable range, not requiring any intervention.

We were unable to restore rScO_2 to baseline values in 20% of patients in the operating room and 12% of patients in ICU. Although the previously reported restoration failure rates for rScO_2 varied between 3% and 20% during CPB [8, 19, 20], none of these studies included patients with deep hypothermic circulatory arrest. This particular sub-group of patients has higher rates of cerebral desaturation that are not amenable to standard interventions during circulatory arrest. Furthermore, unsuccessful reversal of rScO_2 could have potentially resulted in higher number of patients with POD in the interventional group. However, delirium rates in patients with 'successful' vs. 'unsuccessful' reversal vs. 'no interventions', were not significantly different. In addition, patients with poor baseline rScO_2 values could also have contributed to reduced reversibility of rScO_2 decreases, and resulted in reduced numbers of patients receiving successful treatment.

In conclusion, this study suggests that the baseline $\text{rScO}_2 \leq 50\%$ carried better predictive value of POD than a relative change in rScO_2 values during the peri-operative period. It has been previously suggested that low rScO_2 might be a reflection of poor cognitive reserve in awake elderly subjects [25]. It is important to recognise that cognitive reserve is included in most frailty evaluation tools, and the incidence of POD increases exponentially with the advanced frailty scores

[26]. Consequently, baseline rScO₂ might be a quantitative reflection of combined frailty and cognitive reserve scores. Measurements of baseline rScO₂ could optimise peri-operative risk stratification and direct appropriate preventive and therapeutic strategies towards reducing POD.

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